Every Remainder Everywhere

The Commutator Engine: How Fixed Points Disappear in Collatz Inverse Trees

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Abstract

When analyzing the Collatz map backwards, one seeks to prove that predecessors of any seed cover every residue modulo a prime p. The individual inverse operations generate affine transformations $t \mapsto at + b$. When $a \neq 1$, such maps possess an anchor—a fixed point, like $-40°$ in the Celsius-to-Fahrenheit conversion. In modular dynamics, anchors can trap points in small orbits. To guarantee full coverage, one needs an anchorless map: a pure translation $t \mapsto t + d$ with $d \neq 0$, which acts like an unstoppable conveyor belt visiting every hour on the clock of $\mathbb{Z}/p\mathbb{Z}$. This paper explains how the commutator engine pairs a dilation with an affine walk to cancel out the dilation factor completely, producing a transformation with no fixed point and proving that Collatz predecessors hit every remainder everywhere.

1 The Anchor: Why Fahrenheit and Celsius Have a Trap

Consider the familiar formula for converting Celsius to Fahrenheit:

$$F = \frac{9}{5}C + 32.$$

This formula combines two fundamental operations:

1. **A dilation (scaling):** multiplying the input by $\frac{9}{5}$.
2. **A translation (shift):** adding 32.

Whenever the multiplier is not 1 (here it is 1.8), there is always an *anchor point*—a fixed point where the input and output agree. Setting $F = C$:

$$C = \frac{9}{5}C + 32 \implies -\frac{4}{5}C = 32 \implies C = -40.$$

At $-40°$, Celsius and Fahrenheit are identical.

In general, every affine transformation $T(x) = ax + b$ with slope $a \neq 1$ has a unique fixed point:

$$x^* = \frac{b}{1 - a}.$$

Geometrically, the transformation does not move the whole line uniformly; it merely expands or contracts the world away from its anchor at x^*.

2 The Modular Trap: Trapped by an Anchor

In the Collatz inverse tree, we follow numbers backward from a target w. The inverse branches consist of:

- A **right branch** $R(x)$, which multiplies by 4 or 16 (halving steps in reverse).

- A **left branch** $L(x)$, which inverts an odd step $3x+1$ via exact division.

When we reduce these operations modulo a prime $p \geq 5$, every step acts on residues $t \in \mathbb{Z}/p\mathbb{Z}$ as an affine transformation:

$$t \mapsto a \cdot t + b \pmod{p}.$$

Because p is prime, if the multiplier $a \not\equiv 1 \pmod{p}$, the equation has a modular fixed point:

$$t^* \equiv \frac{b}{1-a} \pmod{p}.$$

Why this is a problem. If a transformation has an anchor t^*, it cannot easily visit every residue. Points can get stuck in small orbits or subgroups orbiting around t^*, leaving large parts of $\mathbb{Z}/p\mathbb{Z}$ completely untouched. As long as our backward walks retain their "$-40°$" anchors, we cannot guarantee that Collatz predecessors cover every possible remainder.

3 The Conveyor Belt: A Transformation with No Fixed Point

What kind of transformation has **no fixed point at all**?

Imagine a temperature scale defined simply by:

$$\text{Scale } B = \text{Scale } A + 5°.$$

Can Scale A ever equal Scale B? Never: $A = A + 5$ has no solution. There is no anchor point anywhere.

On the circular clock of residues $\{0, 1, 2, \ldots, p-1\} = \mathbb{Z}/p\mathbb{Z}$, a transformation with **no fixed point** is a pure, nonzero translation:

$$T(t) = t + d \pmod{p} \quad (d \not\equiv 0).$$

A fixed point would require $t \equiv t + d \pmod{p}$, forcing $d \equiv 0$, which is impossible since $d \neq 0$.

More importantly, because p is a prime, repeatedly applying $+d$ turns the transformation into a single, relentless conveyor belt:

$$t_0, \quad t_0 + d, \quad t_0 + 2d, \quad \ldots, \quad t_0 + (p-1)d \pmod{p}.$$

Starting from *any* seed, this sequence visits **every single hour on the clock** before returning to the start.

> **The Core Strategy:** If the loops in our inverse tree can generate even *one* closed-walk transformation with **no fixed point**, that transformation alone forces the set of available residues to become all of $\mathbb{Z}/p\mathbb{Z}$.

4 The Commutator Engine: Erasing the Anchor

Every single edge in the inverse tree has a dilation factor $a \neq 1$. Every basic move has an anchor. How can we construct an anchorless map ($a = 1$) out of components that all have anchors ($a \neq 1$)?

We use a **commutator**. In group theory, when two operations do not commute, their commutator:

$$[A, B] = A^{-1} \circ B^{-1} \circ A \circ B$$

measures the failure of commutativity.

In the 3-adic inverse tree, we construct two specific closed walks at the base class 8:

1. A **pure dilation** $B = (\lambda, 0)$ with $\lambda \not\equiv 0, 1 \pmod{p}$ (proved to exist via the 3-adic loop structure).

2. A **general affine walk** $A = (\alpha, \beta)$ with $\alpha \neq 0$ and a nonzero translation part $\beta \neq 0$ (obtained by taking a left edge and returning via right edges).

Now we form their commutator $[A, B] = A^{-1} \circ B^{-1} \circ A \circ B$.

1. What happens to the dilation? Because the linear slopes multiply, the new slope is:

$$\text{New Slope} = \alpha^{-1} \cdot \lambda^{-1} \cdot \alpha \cdot \lambda = \mathbf{1}.$$

The stretching factors cancel out completely. The multiplier becomes exactly 1.

2. What happens to the translation? Because affine transformations do not commute, the shifts do not cancel. Evaluating the composition explicitly:

$$\begin{aligned} t &\xrightarrow{B} \lambda t \\ &\xrightarrow{A} \alpha(\lambda t) + \beta \\ &\xrightarrow{B^{-1}} \lambda^{-1}(\alpha\lambda t + \beta) = \alpha t + \lambda^{-1}\beta \\ &\xrightarrow{A^{-1}} \alpha^{-1}\Big((\alpha t + \lambda^{-1}\beta) - \beta\Big) = t + \alpha^{-1}(\lambda^{-1} - 1)\beta. \end{aligned}$$

Under the forward convention used in the formalization ($f \circ g = g(f(t))$), this evaluates directly to:

$$[A, B](t) = t + (\lambda - 1)\beta \pmod{p}.$$

Since $\lambda \not\equiv 1$ and $\beta \not\equiv 0$ in the field $\mathbb{Z}/p\mathbb{Z}$, the net shift is:

$$d = (\lambda - 1)\beta \not\equiv 0 \pmod{p}.$$

The commutator has eliminated the anchor point entirely, leaving behind a pure translation:

$$[A, B] = (1, d) \quad \text{with } d \neq 0.$$

5 Closing the Argument: Full Coverage Everywhere

With the anchor destroyed, the proof of modular reachability falls out in three short steps:

1. **Fibre Rigidity:** Let $F_K(c) \subseteq \mathbb{Z}/p\mathbb{Z}$ be the set of residues mod p attained by Collatz predecessors in 3-adic class c at depth K. As depth increases, $|F_K(8)|$ is non-increasing. Being a non-empty sequence of positive integers bounded by p, its size must freeze at a constant $\sigma \geq 1$. At this stable depth, every closed walk acts on the fibre as a bijection.

2. **The Fibre is Everything:** Since the commutator $[A, B] = (1, d)$ is the label of a valid closed walk, it maps the stabilized fibre $F_K(8)$ onto itself:

$$t \in F_K(8) \implies t + d \in F_K(8).$$

Starting from any element $t_0 \in F_K(8)$, repeatedly adding d visits all p residues in $\mathbb{Z}/p\mathbb{Z}$. Therefore:

$$F_K(8) = \mathbb{Z}/p\mathbb{Z} \implies \sigma = p.$$

Since all valid classes at depth K have the same stabilized size σ, *every valid class covers every residue modulo p.*

3. **Exceeding the Bound:** To find a predecessor $n > B$ reaching our target w with $n \equiv r \pmod{p}$, we simply pick a class c at a depth where $3^{K+1} > B$. Because the fibre over c is all of $\mathbb{Z}/p\mathbb{Z}$, there exists a genuine descendant n in that class with $n \equiv r \pmod{p}$. Since $n \geq c > B$ and the abstract tree embeds directly into the genuine Collatz graph, n reaches w.

6 Summary

The reason Collatz predecessors do not get trapped mod p is that the inverse tree is algebraically rich enough to contain non-commuting symmetries:

Individual walks	$\Longrightarrow$	Dilations with anchors	$\Longrightarrow$	Trapped orbits possible
Commutator $[A, B]$	$\Longrightarrow$	Pure translation $(1, d)$	$\Longrightarrow$	No anchor (no $-40°$)
No anchor on $\mathbb{Z}/p\mathbb{Z}$	$\Longrightarrow$	Full cycle of length p	$\Longrightarrow$	**Every remainder everywhere.**